

Zachary J. Levesque Ottawa, ON zlevesque93@gmail.com linkedin.com/in/zachary-levesque github.com/Zachary-Levesque zacharylevesque.site			
TECHNICAL SKILLS	Programming: Python, C, C++, C#, Java, Assembly, Verilog, TypeScript, JavaScript, HTML, MATLAB AI & ML: Neural Networks, PyTorch, TensorFlow, Qiskit, training/validation/deployment, quantum computing Simulation: CAD, TCAD, FEM, Simulink, Multisim, mathematical modeling Circuits & IC Design: Digital + analog design, synthesis (Verilog, Synopsys), signal processing Hardware & Embedded Systems: Arduino, STM32, ESP (ESP-IDF), R-Pi, optical linecards, I2C/SPI/UART protocols Software & Tools: FastAPI, React.js, PostgreSQL, Docker, Git, Linux Professional: Data analysis, problem solving, leadership, bilingual (French, English)		
EDUCATION	University of Ottawa Ottawa, ON <i>B.Sc./B.A.Sc. (Double Major) in Electrical Engineering and Physics</i> Sep 2022 – Expected Apr 2027 - CGPA: 9.92/10 (4.0 GPA) 8× Dean's List (2022–2026); multiple Faculty of Engineering & Science Recognitions of Excellence - Awarded the International Experience Bursary to attend NCKU Digital IC Design & Synthesis (Taiwan, 2025): 98% best in class - ASCOT Recognition of Excellence Scholarship, Electro Federation Canada (2024) - Admission Scholarship, University of Ottawa (2022) – awarded for a 98% average (4.0 GPA)		
EXPERIENCE	Ranovus Inc. Kanata, ON <i>Systems Verification Engineer</i> May 2026 – Present - Design and develop embedded firmware and board-level communication protocol stacks (I2C, SPI) for MCU-based hardware platforms - Perform PCB-level signal analysis and hardware debugging, backed by automated Python test scripts and validation infrastructure - Diagnose and resolve issues across the hardware/firmware boundary through hardware configuration and low-level debugging Seets Inc. Ottawa, ON <i>Co-Founder & CEO</i> May 2026 – Present - Co-founded and lead an IoT occupancy-tracking startup, owning product vision, system architecture, and full-stack engineering execution - Took the product from capstone concept to a live pilot deployment at the University of Ottawa Heart Institute - Building a scalable solution at the intersection of hardware, software, and real-world analytics APRIQuOt Ottawa, ON <i>Quantum AI Researcher (Intern)</i> Sep 2025 – Apr 2026 - Applied machine learning to quantum state tomography, training neural networks to reconstruct multi-qubit quantum states - Used Qiskit to simulate and study quantum chaos in complex multi-qubit systems, supporting ongoing quantum information research - Implemented and optimized algorithms to improve reconstruction accuracy University of Ottawa Ottawa, ON <i>Teaching Assistant – ELG 3525, PHY 1722</i> Sep 2025 – May 2026 - Led tutorials across two courses for both engineering (Signal & System Analysis) and physics (Physics II) undergraduates - Provided 1:1 and small-group academic support to students through the Physics Help Centre - Helped students strengthen problem-solving strategies and build confidence in core technical concepts Huawei Ottawa, ON <i>Systems Engineer – Optical Communications (Intern)</i> May 2025 – Aug 2025 - Conducted R&D on high-speed optical data transmission, designing communication links using EDFA, WSS, transceivers, and OSA hardware - Built Python automation for link control and data acquisition, applying statistical analysis to boost throughput and measurement turnaround - Applied mathematical models and PyTorch CNNs to boost SNR from 20 to 40 dB, selected for production over 5 approaches Thin-Film Electronics Laboratory Ottawa, ON <i>Semiconductor Engineer (Intern)</i> Jan 2025 – Apr 2025 - Conducted theoretical research on non-linear charge injection behavior in n-type organic transistors - Developed TCAD physical simulation models to characterize and analyze thin-film transistor parameters - Built Python and MATLAB tools to analyze and improve transistor performance and efficiency Hydro Ottawa Ottawa, ON <i>Data Engineering Intern</i> May 2024 – Aug 2024 - Analyzed GIS data for the city's electrical distribution network to guide infrastructure planning and development - Built ML models to automate crew placement, material and tool allocation, project tracking, and budget forecasting - Developed a Java program to transfer data between software applications, improving workflow efficiency and integration <tr> <td>PROJECTS</td><td> Knovara <i>FastAPI, ChromaDB, OpenAI API, Next.js, TypeScript</i> May 2026 – Present - Architecting an AI onboarding copilot with a RAG ingestion pipeline (FastAPI + ChromaDB), piloting with Seets as first customer Smart Flood Detection System <i>C, ESP-IDF, ESP32, I2C</i> Jul 2026 – Aug 2026 - Built and deployed a live ESP32-based IoT flood detection system with I2C water-level sensing and WiFi Telegram alerting Mini Tensor Engine <i>C++, AVX2 SIMD, Python</i> Jan 2026 – May 2026 - Built a C++ inference runtime with AVX2-vectorized, tiled matmul, achieving a 3.37× speedup at 1024³ Adaptive Market Regime Framework <i>Python, HMMs, RL</i> Aug 2025 – Dec 2025 - Designed an HMM-based regime detection system with RL-driven allocation, achieving a 1.09 Sharpe ratio vs. 0.95 for SPY Real-Time Trading System <i>FastAPI, React, PostgreSQL, Redis</i> Jan 2025 – May 2025 - Architected a full-stack trading platform ingesting live market data and generating explainable buy/hold/sell signals CircuitFlow <i>TypeScript, Circuit Design & Analysis</i> Sep 2022 – Dec 2022 - Built a real-time circuit simulator using matrix-based nodal analysis for DC/AC/transient circuit analysis Additional Projects: ParaPicture Solutions, Sports Strategy Engine, AI Games & Logic Solvers, AI Agent/Workflows, Quantum Math Quest, 1Chat4You </td></tr>	PROJECTS	Knovara <i>FastAPI, ChromaDB, OpenAI API, Next.js, TypeScript</i> May 2026 – Present - Architecting an AI onboarding copilot with a RAG ingestion pipeline (FastAPI + ChromaDB), piloting with Seets as first customer Smart Flood Detection System <i>C, ESP-IDF, ESP32, I2C</i> Jul 2026 – Aug 2026 - Built and deployed a live ESP32-based IoT flood detection system with I2C water-level sensing and WiFi Telegram alerting Mini Tensor Engine <i>C++, AVX2 SIMD, Python</i> Jan 2026 – May 2026 - 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